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Problem Solving Protocol

Identification of significant gene expression changes in multiple perturbation experiments using knockoffs

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Abstract

Large-scale multiple perturbation experiments have the potential to reveal a more detailed understanding of the molecular pathways that respond to genetic and environmental changes. A key question in these studies is which gene expression changes are important for the response to the perturbation. This problem is challenging because (i) the functional form of the nonlinear relationship between gene expression and the perturbation is unknown and (ii) identification of the most important genes is a high-dimensional variable selection problem. To deal with these challenges, we present here a method based on the model-X knockoffs framework and Deep Neural Networks to identify significant gene expression changes in multiple perturbation experiments. This approach makes no assumptions on the functional form of the dependence between the responses and the perturbations and it enjoys finite sample false discovery rate control for the selected set of important gene expression responses. We apply this approach to the Library of Integrated Network-Based Cellular Signature data sets which is a National Institutes of Health Common Fund program that catalogs how human cells globally respond to chemical, genetic and disease perturbations. We identified important genes whose expression is directly modulated in response to perturbation with anthracycline, vorinostat, trichostatin-a, geldanamycin and sirolimus. We compare the set of important genes that respond to these small molecules to identify co-responsive pathways. Identification of which genes respond to specific perturbation stressors can provide better understanding of the underlying mechanisms of disease and advance the identification of new drug targets.

Keywords: deep learning, genomics, gene expression, chemoinformatics, knockoffs

INTRODUCTION

The elucidation of the genetic and biochemical mechanisms underlying cellular function requires perturbation of the system [1]. Perturbation approaches have been successful in understanding fundamental pathways in yeast [2], humans [3] and other organisms [4]. Perturbation experiments also provide a deeper understanding of the mechanism of action of small molecule compounds and guide opportunities for drug repurposing [5] and repositioning [6–9].

A compendium of expression profiles corresponding to genetic and environmental perturbations can be used to functionally characterize unknown genes and to characterize pharmacological small molecules [10]. A match between the gene expression pattern of a strain with a deletion in an unknown open-reading frame and a strain with a knockout in a known gene reveals cellular function because mutants that display similar expression profiles are likely to share cellular functions. Similarly, an expression pattern match between an uncharacterized pharmacological agent and a knockout strain or another well-characterized agent reveals the potential mechanisms of action of the uncharacterized agent.

Hierarchical clustering (HC) has been used to match the expression patterns [10]. However, HC suffers from several drawbacks including that it cannot detect multiple components within the response to a given perturbation [6]. To deal with this challenge, CMap uses a nonparametric rank-based pattern-matching strategy via Kolmogorov–Smirnov (KS) statistics [6]. The key idea of CMap is to find the connectivity between disease gene expression signatures and compound-induced gene expression profiles based on rankings of genes in a disease signature within the drug-perturbed expression profiles. This method assesses the similarity of a 'query signature' to the reference expression profiles in the data set. For example, a query signature could be any list of genes with expressions correlated with a subtype of diseases. One limitation of the CMap method is that it cannot test the statistical significance of the connections observed [6]. To correct this defficiency, one proposal suggested a more robust method which allows for testing of the statistical significance of the connections [11]. EXtreme sum (XSum), eXtreme cosine (XCos), Pearson correlation (XCor) and Spearman correlation (XSpe) have been proposed as measures of gene expression profile

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similarity to extend CMap [12, 13]. Systematic evaluations and reviews for CMap-related methodologies have been performed [14, 15], and those evaluations have been extended to the L1000 data set that contains the expression profiles of multiple human cell lines and various types of perturbagens [16].

Although CMap-related methodologies has proven to be beneficial for drug discovery and development and helpful for generation of novel hypotheses beyond classical pharmacology [17, 18], they also come with some limitations [19]. The main challenge is that rank-based methods such as CMap can suffer from high False Discovery Rate (FDR), which quantifies the expected proportion of discoveries which are falsely rejected. A drug may perturb the expression of anywhere from thousands of genes to only a few genes. Thus, for ranking-based method, it is hard to select a threshold N to determine the set of top N genes as drug-perturbed genes. Furthermore, ranking-based or correlation-based methods may also miss potential drugs as ranking and correlation only captures a limited proportion of the information in gene expression profiles. Finally, there is a lack of rigorous quantitative method that can control the FDR for computational predictions.

This previous work highlights two challenges with functional discovery by compendium pattern matching—many genes respond to the perturbation not because they are directly related to the perturbation, but because they are responding to changes in expression of other genes to maintain cellular homeostasis. These second-order effect genes are not directly useful for the characterization of the unknown gene or the pharmacological agent and represent noise in the classification process. A second challenge is that gene expression changes to genetic and environmental perturbations can be highly nonlinear. When yeast cells are presented with a small molecular dihydrofolate reductase inhibitor 5-FU, the expression of drug efflux pumps is dramatically increased to remove the molecule from the cell. It is not known, a-priori, what the functional form of the change in expression will be in response to a perturbation concentration.

There is a need for a rigorous statistical modelling method that goes beyond ranked correlation information and can achieve high classifying power of drug–disease relationships while controlling the FDR. The FDR metric has been popular in a variety of domains such as the discovery of eQTLs, Bayesian association tests for rare variation in autism spectrum disorder studies [20–22] and identification of putative causal loci in whole genome sequencing data [23]. The knockoff method accomplishes variable selection with rigorous FDR control under arbitrary correlation structure without strong modelling assumption constraints [24]. Knockoffs have been applied to a wide variety of fields such as random design settings [24], whole-genome sequencing data [23] and genome-wide association studies [25]. Note that knockoffs are a statistical method for feature selection; the term does not refer to a genetic mutation to remove the DNA for a genetic region which is termed as knockout.

In this work, we propose a statistical protocol based on model-X knockoffs framework [24] to identify significant drug-perturbed gene expressions in multiple perturbation experiments while controlling FDR. It is a general method since it makes no assumptions on the functional form of the dependence between the gene expression changes and the perturbations and provides finite sample FDR control for the set of important gene expression profiles. It is a flexible protocol that can incorporate deep learning for feature selection [26, 27] and classic methods such as Lasso regression [28] genes that respond directly to the perturbation. The main idea of knockoffs for perturbation-gene expression data is that it first constructs synthetic gene expression data that mimic the true gene expression data in terms of the correlation structure but is conditionally independent of different perturbation conditions given the true gene expression data. The knockoff gene expression data will serve as negative controls to help us identify the truly important drug-perturbed gene expression profiles while controlling the FDR at a pre-specified target level.

We demonstrate this problem-solving protocol by conducting a functional analysis and characterization of several small molecule agents from the (LINCS). The LINCS program is an NIH Common Fund program led by over 15 institutions that catalogs how human cells react to chemical, genetic and disease perturbations [29, 30]. A fundamental difficulty and challenge in drug repositioning is to establish the relation between the mechanisms of diseases and the role of small-molecule therapeutics. Using gene expression profiling approach to catalog the biological responses to a large number of perturbations is not entirely new [6, 31]. The LINCS program has been implemented in two phases and the first phase provides publicly available data about gene and protein expression changes induced by a variety of perturbations. A comprehensive description of the LINCS program and computational methods based on LINCS program in pharmacogenomics in the literature to date can be found in [15, 30].

METHODS

All previous work has used knockoffs for feature selection—that is, to identify the genetic basis of phenotypic differences. In statistical terms, feature selection is the process of selecting a subset of relevant features given the response variable. In our paper, we are performing a response variable selection task which is motivated by our application. In our work, the perturbation conditions are pre-specified when setting up the experiments and we observe the gene expression changes under different perturbations. Thus, the gene expressions are considered as response variables (dependent variables), and the perturbations are considered as features (independent variables). In this application, we hope to identify the gene expressions (response variables) that are directly related to the perturbation conditions, which is a *response variable selection* problem. For a selection rule that selects a subset of the responses, Figure 2 shows the relationship between the feature selection problem and the response selection problem.

We consider genes that directly respond to the drug perturbations to be *important* and our objective is to select these responses. In order to accomplish this objective, we describe our proposed method denoted as model-Y knockoffs based on model-X knockoffs [24]. The traditional Bonferroni method for family-wise error rate control is too conservative for thousands of simultaneous hypothesis tests and may lead to many missed findings. We provide a proof that model-Y knockoffs gives finite sample FDR control enabling this protocol to able to identify as many significant genes as possible while maintaining a low false positive rate. We present an algorithm for generating and using model-Y knockoffs in a manner analogous to model-X knockoffs to perform response (drug-perturbed gene expressions) selection to identify the truly significant drug-perturbed genes. Finally, we give a non-trivial method for employing model-Y knockoffs for response variable selection in (DNN)s. A flow chart of model Y knockoffs methodology is provided in Figure 1.

Problem statement

Let X_i encode the ith perturbation and let Y_i encode the gene expression measurement vector in *response* to the ith perturbation. Assume we have n i.i.d. random variables (X_i, Y_i), where

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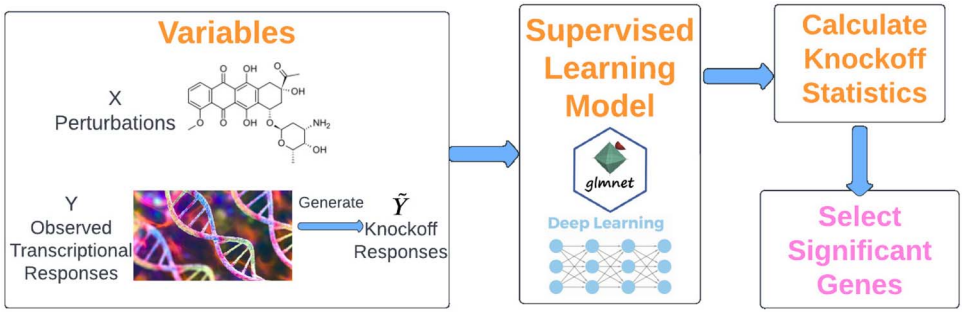


Figure 1. Flow chart of the model-Y knockoffs methodology.

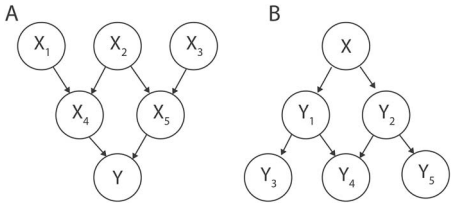


Figure 2. (**A**) Feature selection problem. The goal is to identify features X_4 and X_5 as important because $Y \perp\!\!\!\perp \{X_1, X_2, X_3\} | \{X_4, X_5\}$. The response Y only depends on $\{X_1, X_2, X_3\}$ through the important features $\{X_4, X_5\}$. (**B**) Response selection problem. The goal is to identify response variables Y_1 and Y_2 as important because $X \perp\!\!\!\perp \{Y_3, Y_4, Y_5\} | \{Y_1, Y_2\}$. The perturbation X only affects $\{Y_3, Y_4, Y_5\}$ through the important responses $\{Y_1, Y_2\}$.

$X_i \in \mathbb{R}^p$, $Y_i \in \mathbb{R}^r$ assembled into two data matrices $\mathbf{X} \in \mathbb{R}^{n\times p}$ and $\mathbf{Y} \in \mathbb{R}^{n\times r}$ such that the ith row of $\mathbf{X}$ is X_i and the ith row of $\mathbf{Y}$ is Y_i. For example, in multiple perturbation experiments, X_i is an indicator vector for the perturbation over p possible perturbations, and Y_i is the r-dimensional transcriptional profile. A response variable Y_j for $j \in \{1, \ldots, r\}$ is said to be unimportant if and only if Y_j is independent of X conditionally on the other responses Y_{-j}, where $Y_{-j} = \{Y_1, \ldots, Y_r\} \backslash \{Y_j\}$. The set of unimportant variable indices is denoted by $\mathcal{H}_0$ and we call a variable Y_j important if $j \notin \mathcal{H}_0$. The set of the important variable indices is denoted as $\mathcal{S}$. The unimportant responses are conditionally independent of the covariates given the important responses $\{Y_j\}_{j\in\mathcal{H}_0} \perp\!\!\!\perp X \mid \{Y_j\}_{j\in\mathcal{S}}$. Our goal is to identify as many important responses as possible while keeping the false discovery rate (FDR) under control, where $\text{FDR} = \mathbb{E}\left[|\hat{\mathcal{S}} \cap \mathcal{H}_0|/(|\hat{\mathcal{S}}| \vee 1)\right]$ and $\hat{\mathcal{S}}$ denotes a selected subset of the important response variable indices.

Model-Y knockoffs

The key idea to adapting model-X knockoffs for response variable selection is the observation that the generation and use of model-X knockoffs only depends on the joint distribution between the explanatory variables and the response F_{XY}. Model-X knockoffs assume that the marginal distribution F_X is known exactly and this allows the conditional distribution $F_{Y|X}$ to be unspecified. In response variable selection problems, we have abundant information about the marginal distribution F_Y, so we use the fact that the joint distribution can be factorized as $F_{XY} = F_{X|Y}F_Y$. This leads to a symmetry that carries through the theory that we exploit in our development of model-Y knockoffs.

Definition 1. (Model-Y knockoffs). A random vector $\tilde{Y} = (\tilde{Y}_1, \ldots, \tilde{Y}_r)$ is the model-Y knockoffs of a random vector $Y = (Y_1, \ldots, Y_r)$ if it satisfies two properties: (i) Pairwise exchangeability: for any subset $\mathcal{D} \subset [r]$, $(Y, \tilde{Y})_{\text{swap}(\mathcal{D})} \overset{\text{d}}{=} (Y, \tilde{Y})$, i.e. swapping the entries Y_j and $\tilde{Y}_j$ for each $j \in \mathcal{D}$ leaves the joint distribution invariant and (ii) Conditional independence: $\tilde{Y} \perp\!\!\!\perp X \mid Y$, i.e. $\tilde{Y}$ is independent of features X given the response Y, which can be guaranteed by constructing $\tilde{Y}$ without looking at X.

We provide our theorem to establish the finite sample FDR control for response variables.

Theorem 1.1. (Model-Y knockoffs). Let $W_j = w_j([\mathbf{Y}, \tilde{\mathbf{Y}}], \mathbf{X})$ be the knockoff statistic to measure the importance for each Y_j, $j \in [r]$. Here, functions $w_j, j \in [r]$ are required to satisfy the following flip-sign property:

$$w_j\left([\mathbf{Y}, \tilde{\mathbf{Y}}]_{\text{swap}(\mathcal{D})}, \mathbf{X}\right) = \begin{cases} w_j([\mathbf{Y}, \tilde{\mathbf{Y}}], \mathbf{X}), & j \notin \mathcal{D}, \\ -w_j([\mathbf{Y}, \tilde{\mathbf{Y}}], \mathbf{X}), & j \in \mathcal{D}, \end{cases} \tag{1}$$

where $\mathcal{D}$ is defined in Definition 1. For any $q \in [0, 1]$, let

$$\tau := \min\left\{t > 0 : \frac{1 + |\{j : W_j \le -t\}|}{|\{j : W_j \ge t\}|} \le q\right\}. \tag{2}$$

Then, we can control the $\text{FDR} = \mathbb{E}\left[|\hat{\mathcal{S}} \cap \mathcal{H}_0|/(|\hat{\mathcal{S}}| \vee 1)\right]$ at the target level of q by selecting importance responses

$$\hat{\mathcal{S}} = \{j : W_j \ge \tau\}.$$

Proof. The proof of the theorem follows similarly to model-X knockoffs by Candes *et al.* [24]. ■

Algorithm

The proof of Theorem 1 shows that swapping the roles of X and Y yields valid knockoffs for response variable selection. This fact means that we can switch the roles of X and Y while fitting a model and perform response selection. In the causal inference context, it is not generally appropriate to swap the features with the responses. However, in the context of response selection, we can swap X and Y while using model-X knockoffs. Once we swap X and Y, the original response variables Y becomes the features in the swapped model. We summarize the key steps in Algorithm 1.

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Algorithm 1. Response Selection using Knockoffs

1: Generate the knockoffs, $\tilde{Y}$, of the original response variable, Y, such that

$$(\tilde{Y}, Y) \sim \mathcal{N}(\mathbf{0}, \mathbf{G}), \sim \text{s.t.} \sim \mathbf{G} = \begin{pmatrix} \boldsymbol{\Sigma} & \boldsymbol{\Sigma} - \text{diag}\{\mathbf{s}\} \\ \boldsymbol{\Sigma} - \text{diag}\{\mathbf{s}\} & \boldsymbol{\Sigma} \end{pmatrix},$$

where $\mathcal{N}(\mathbf{0}, \mathbf{G})$ is a multivariate Gaussian distribution function with covariance $\mathbf{G}$, $\boldsymbol{\Sigma}$ is the covariance within the knockoff and original variables, and $\boldsymbol{\Sigma} - \text{diag}\{\mathbf{g}\}$ is the covariance between knockoff and original response variables. By construction, knockoff variables make s such that the correlation between original and knockoff variables is as close to zero as possible.

2: **if** A linear relationship is assumed between the perturbation indicator, X, and the original response, Y **then**

3: Construct the response importance measure and knockoff statistics using Lasso with Model-Y Knockoffs (Lasso-MYK)

4: **else**

5: Construct the response importance measure and knockoff statistics using DNNs with Model-Y Knockoffs (DNN-MYK)

6: **end if**

7: Filter the important response using the threshold τ in Equation 2 with the target FDR level q.

Generalized linear Model-Y knockoffs

To perform response selection under a generalized linear model setting, we fit a regularized multinomial logistic regression model using Lasso [28] with the original response Y augmented with its knockoffs $\tilde{Y}$ in the role of the covariate variables and the perturbation indicators X in the role of the response variable. The optimal Lasso solution is

$$\hat{\boldsymbol{\beta}}(\lambda) \in \underset{\boldsymbol{\beta} \in \mathbb{R}^{2r \times p}}{\arg\min} \left\{ \frac{1}{2} \|\mathbf{X} - [\mathbf{Y}, \tilde{\mathbf{Y}}]\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}. \tag{3}$$

We define $Z_j = \sup\left\{\lambda : \hat{\boldsymbol{\beta}}_j(\lambda) \neq 0\right\}$ to be the point, λ, on the Lasso path at which the response Y_j first enters the model. Then, we compute the standard knockoff statistics $W_j = Z_j - \tilde{Z}_j$, which is likely to be large for positive important responses and negative for null responses. The set of important responses, $\hat{\mathcal{S}}$, is be selected with controlled FDR using Step 3 as described previously. This Lasso-based procedure can be extended to other generalized linear models.

Nonlinear model-Y knockoffs

If a nonlinear relationship between Y and X is more appropriate for the data set of interest, deep neural networks DNNs can be used to construct the knockoff statistics and perform response selection. But, it is non-trivial to construct a knockoff statistic that satisfies the flip-sign property within a deep neural network (DNN). We introduce a pairwise-competing layer as the first layer of a DNN where the inputs are both the response Y and its knockoffs $\tilde{Y}$. In the pairwise-competing layer, there are r nodes and the jth node connects only response Y_j and its knockoff $\tilde{Y}_j$ to encourage competition between the two corresponding weights ψ_j and $\tilde{\psi}_j$. This technique is inspired by Lu *et al.* [26] who used it for the purpose of increasing the interpretability and reproducibility of DNNs. We denote the weights from the competing layer to the next layer as θ_0, the weights for the last layer to the output X as θ_L and the weights for each pair of connected intermediate layers as $\theta_1, \theta_2, \ldots, \theta_{L-1}$. We define $\Theta_j = \theta_0 \odot (\theta_1 \cdot \theta_2 \cdots \theta_L)$, where $\odot$ represents entry-wise matrix multiplication. Here, $(\theta_1 \cdot \theta_2 \cdots \theta_L)$ represents the matrix multiplication for the weights of all connected layers and can be interpreted as the importance of the Y_j in the multilayer perceptron. Thus, we define the response importance Z_j as $\psi_j \times \Theta_j$, its knockoff importance $\tilde{Z}_j$ as $\tilde{\psi}_j \times \Theta_j$ and we obtain the model-Y knockoff statistics $W_j = Z_j - \tilde{Z}_j$, where $j \in [r]$. As in the case of the Lasso, these definitions are sufficient to employ Theorem 1.

SYNTHETIC DATA EXPERIMENTS

These synthetic data experiments test whether our theoretical proofs regarding finite-sample FDR control hold empirically. We characterize the response variable selection performance for both linear and nonlinear models by exploring the effects of the key data set properties including: total number of response variables r, sample size n, correlation ρ between features and sparsity level s in features. We generated synthetic data sets using four mechanisms varying the *model* (linear and nonlinear) and the *features* $\mathbf{X}$ (continuous or binary) shown in Tables S1 and S2 in the Supplementary Materials.

We address four key questions using our simulation experiment data. We demonstrate our approach can identify the important responses with high power while controlling FDR (i) in both a linear and a nonlinear setting with continuous or binary features, (ii) in a high-dimensional setting with $r > n$, (iii) when the features are highly correlated, (vi) when the responses only rely on part of the features and (V) when the number of samples is small.

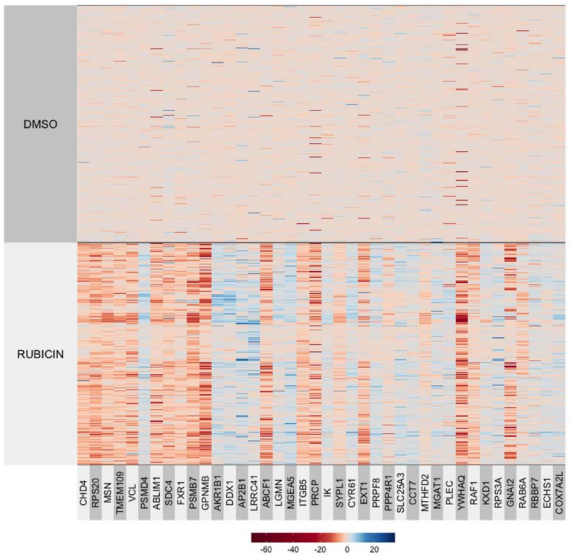
We describe the data generation, implementation details of our neural network architecture, tuning parameter choices and key results in the Supplementary Materials. In this paper, we try to avoid data set specific tuning as much as possible and choose commonly used parameters for DNNs. Code to reproduce these experiments is available at https://github.com/flahertylab/deepYknockoff. We have built an R package and made it available at https://guang-yu-zhu.github.io/YKnock/.

REAL DATA ANALYSIS

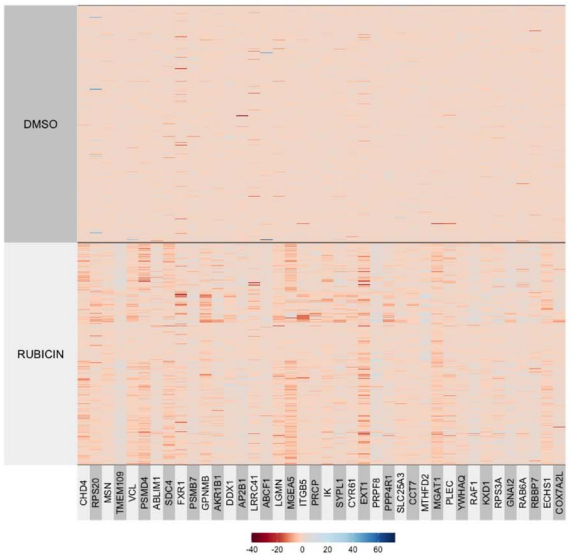
We examined genomic data from the Library of Integrated Network-Based Cellular Signature (LINCS) Phase I L1000 data set which was collected with the aim to improve understanding of human disease and developing new therapies via cataloging how human cells respond to chemical, genetic and disease perturbations [30]. The data set GSE92742 is available online (http://lincsportal.ccs.miami.edu/dcic-portal/) and is described in detail by Subramanian *et al.* [1]. We analyzed a subset of the L1000 Level 4 data dealing with the following small molecule perturbations: anthracycline, vorinostat, trichostatin-a, wortmannin, geldanamycin and sirolimus. The data preprocessing details are described in the Supplementary Materials. To obtain the samples in the treatment group, we obtain 937 samples by subsetting the original data from compound treatments exposed to all anthracycline drug perturbagens (RUBICIN) including 'daunorubicin', 'epirubicin', 'idarubicin', 'doxorubicin' and 'pirarubicin' in L1000 to get the treatment group. We randomly sample 1000 samples with Dimethylsulfoxide (DMSO) perturbations since DMSO is the control for compound treatments considered in L1000.

To map the LINCS data to the notation of the Problem Statement section, the experimental sample/run is indexed by i. For each experimental run, the LINCS consortium data provide a vector, Y_i, of the transcriptional levels of $r = 978$ 'landmark genes'—since we use Level 4 data, each element of Y_i is a z-score.

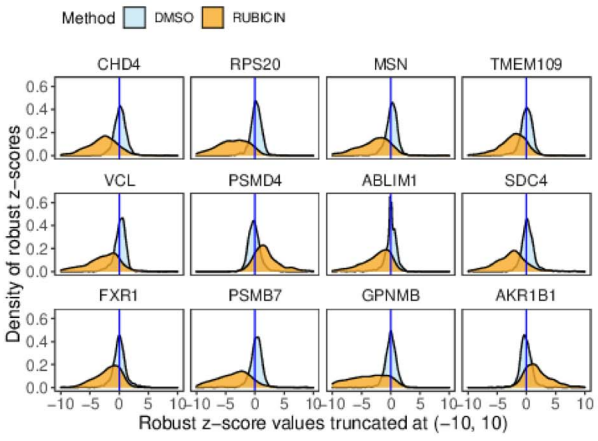
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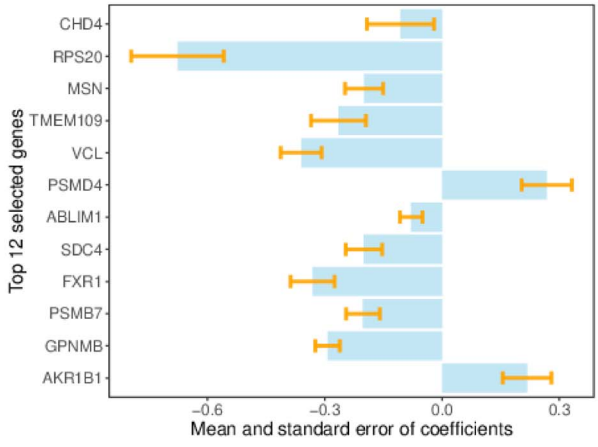
(A) Heatmap of true robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(B) Heatmap of knockoff robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

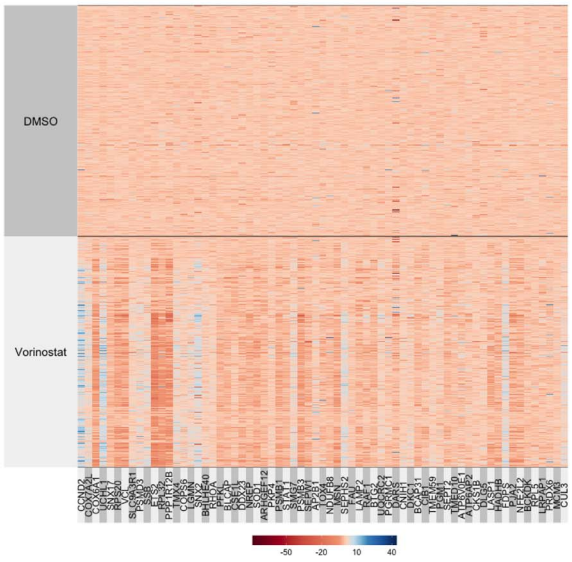
Figure 3. Heatmap of true robust Z-scores for selected genes for anthracycline (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

Since experiment i was done with a specific perturbation to the cells, that is recorded in X_i. The subset of the full LINCS data set contains six perturbations, X_i is length two where at most one element is set to 1 as an indicator of the perturbaton or DMSO control; each perturbation was run in a distinct analysis. The perturbation is then (in our focused subset of the LINCS data) a small molecule perturbation. The samples span variations in cell line, time point and concentration. Of course, if an investigator was interested in responses at a particular time point or cell line, the indicator variable, X_i, can easily be constructed to examine those differences. There is a myriad of possible investigations of the data using this method and our aim here is to demonstrate its capabilities.

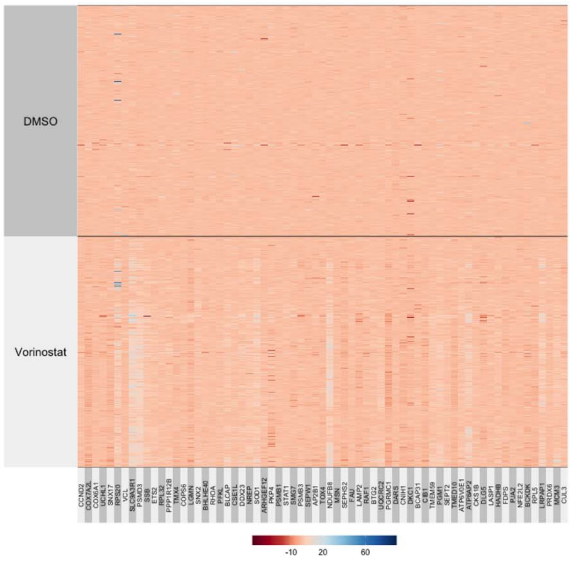
We construct a network diagram to reveal the individual and shared important transcriptional responses genes when perturbed by the selected six drugs. We did not compare our method with CMap-based since they are designed to solve different problems. Our method identifies of important transcriptional changes in response to chemical and genetic perturbations, which is a supervised response selection problem. In contrast, CMap-based methodologies create a reference collection of gene-expression profiles and rank the KS statistics to connect disease gene expression signatures to drug expression profiles, which is a pattern-matching and similarity-testing problem.

We set the target FDR=0.1 and used Lasso-MYK with binary features in a logistic regression model since it is the most robust method as suggested by our simulation studies. We first present our findings for anthracycline drugs. Then, we repeat the same statistical analysis procedure to identify the important genes for vorinostat, trichostatin-a, wortmannin, geldanamycin and sirolimus, respectively. Finally, we construct a network diagram (Figure 9) to show the individual and shared important

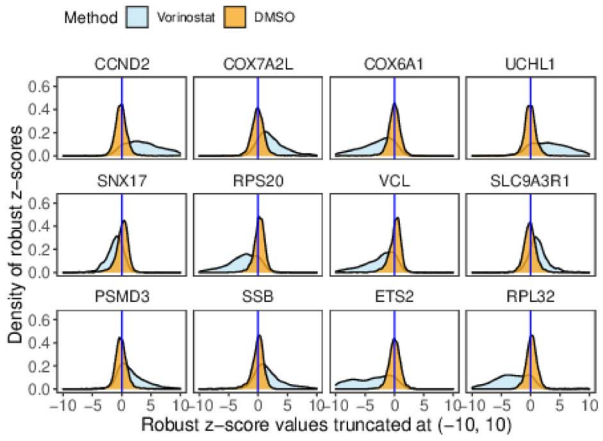
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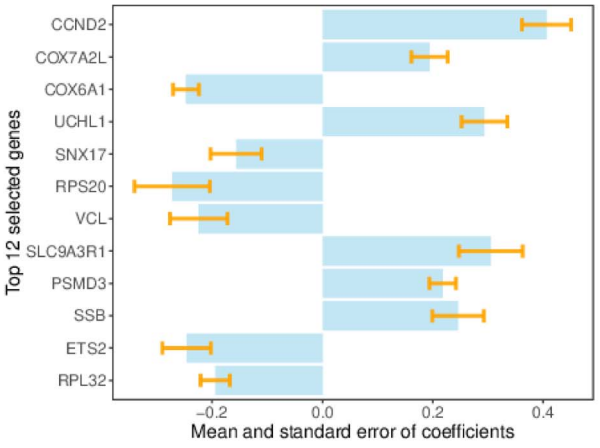
(A) Heatmap of true robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(B) Heatmap of knockoff robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

Figure 4. Heatmap of true robust Z-scores for selected genes for vorinostat (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

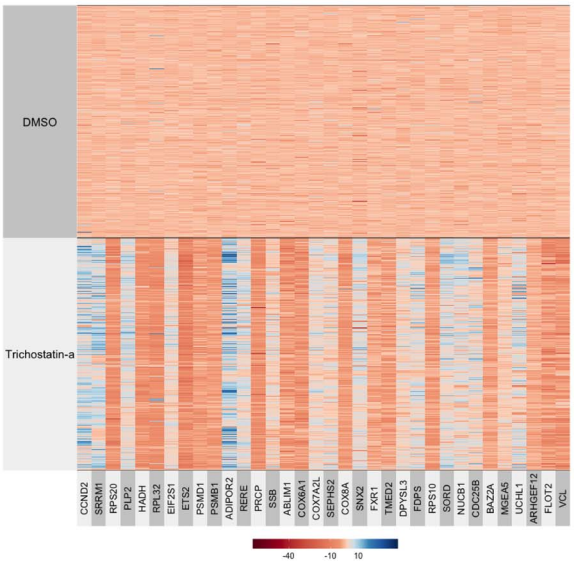
transcriptional responses genes when perturbed by the selected six drugs.

The data set used is a gene expression data set, so the important genes are those that exhibit a transcriptional response to the perturbation conditional on all of the other genes. To validate our selected important genes, we provide the heatmap to demonstrate the important genes with the same randomly selected number of unimportant genes and it is clear that the genes identified by our method carry stronger signals and patterns than the unimportant ones and the unimportant gene group shows little difference between the drug perturbations compared with the DMSO control groups. Check Figures S5–S10 in the Supplementary Materials for details.

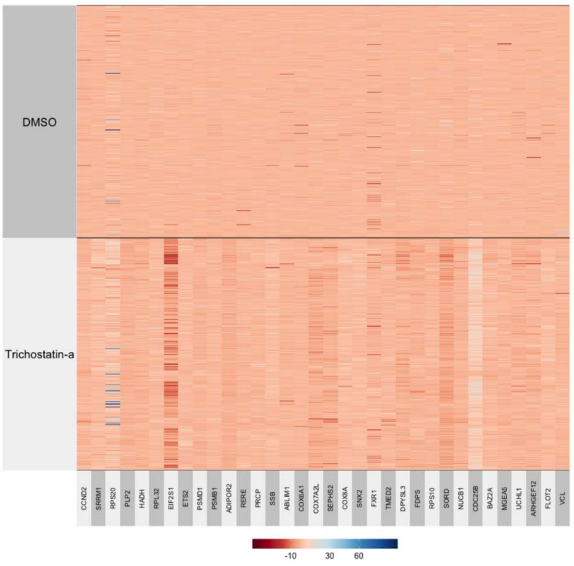
Anthracyclines For anthracycline drug perturbagens (RUBICIN), the model-Y knockoff procedure selected 40 unique landmark genes across all 100 replications. Figure 3(a) shows the robust Z-score of the selected genes under DMSO control and anthracycline treatment perturbation. Clearly, the selected important genes have a different signature under the control and treatment groups. In comparison, we also display the average of the knockoffs, $\tilde{Y}$, across different replications for the selected genes. The model-Y knockoffs, $\tilde{Y}$, do not demonstrate such strong association with the treatment perturbation. Figure 3(c) and 3(d) show density plot of the robust Z-score and bar plots of the mean and standard error of the coefficients for top 12 selected genes across 100 replications based on their selection frequency.

Anthracycline compounds are a widely used class of cancer chemotherapy that primarily act by intercalating with DNA and inhibiting both DNA metabolism and RNA synthesis. Lasso-MYK identified Chromodomain Helicase DNA Binding Protein 4 (*CHD4*) which encodes a DNA-binding helicase protein. A recent study reported that 'CHD4 depletion dramatically decreases tumor-forming behavior of AML cells and modulates expression of genes

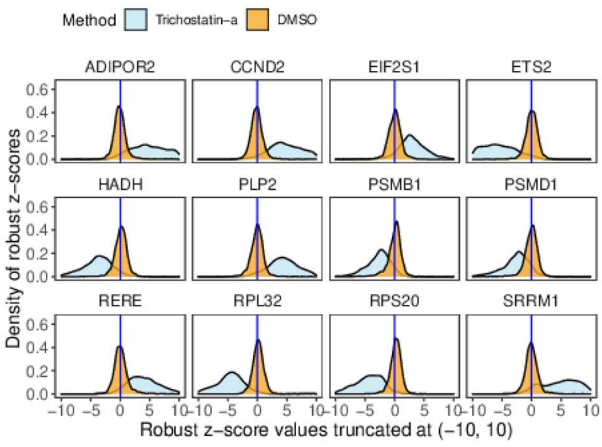
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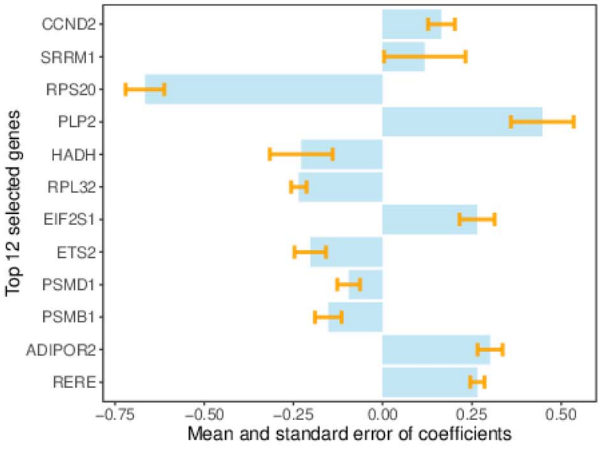
(A) Heatmap of true robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(B) Heatmap of knockoff robust Z-scores for selected important genes at FDR=0.1 across 100 replications.



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



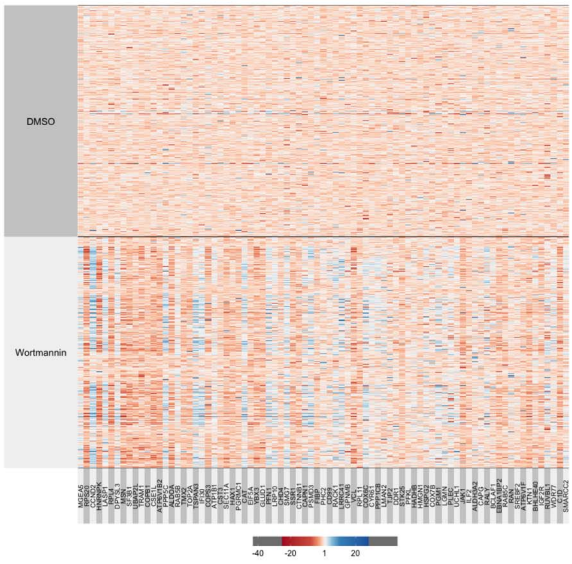
(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

Figure 5. Heatmap of true robust Z-scores for selected genes for trichostatin-a (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

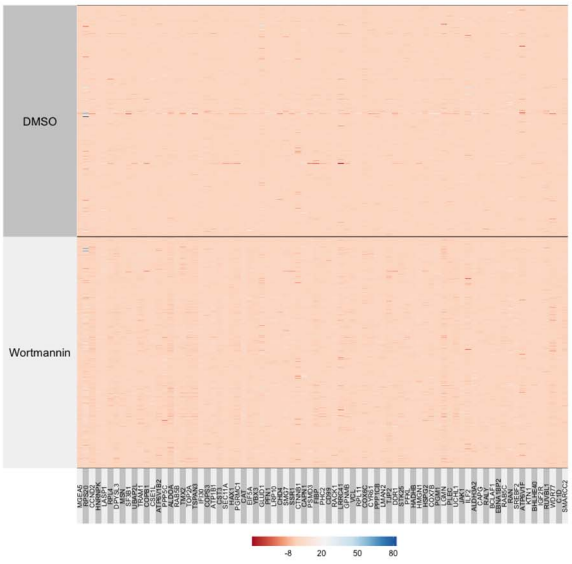
associated with tumor colony formation' [32]. The identification of *CHD4* by our procedure with a negative coefficient suggests that treatment with anthracycline decreases *CHD4* levels and contributes to the anticancer efficacy of the drug by reducing tumor colony formation. Vinculin (*VCL*) is identified as an important response and has a reduced expression after perturbation with anthracycline. Decreased expression of this gene is associated with cardiomyopathy which is a common side effect of anthracycline treatment [33]. Finally, FMR1 Autosomal Homolog 1 (*FXR1*) was recently reported in a study involving triple-negative breast cancer patients [34]. That study reported that a locus on the q arm of chromosome 3, later localized to *FXR1*, is strongly associated with distant metastasis in triple-negative breast cancer. The observation that *FXR1* is an important gene and has a negative coefficient suggesting the hypothesis that anthracycline may have a therapeutic effect for triple-negative breast cancer patients by decreasing expression of *FXR1* and this reduces the likelihood of distant metastasis. Finally, GPNMB is dramatically downregulated in response to anthracycline. A recent study showed that '...cells treated with macrophage-produced GPNMB showed higher resistance to the cytotoxic effect of doxorubicin'[35]. The direct effect of doxorubcin is a downregulation of GPNMB (our data) and doxorubicin is a toxic perturbation, so it is a logical inference that increasing the levels of GPNMB leads to the cytotoxic effects.

Vorinostat The negative coefficient of *COXA1* indicates that vorinostat has treatment effect on Hurthle cell neoplasms (Figure 4). It has been reported that Cytochrome C Oxidase Subunit 6A1 (*COXA1*) is involved in mitochondrial metabolism and is significantly overexpressed in tumors in Hurthle cell neoplasms versus normal tissue [36]. D'Arcy and Linder [37] reported that Ubiquitin C-Terminal Hydrolase L1 (*UCHL1*) showed decreased activity in the tumor tissues, potentially indicating a tumor-suppressive role for this protein. It has also been reported

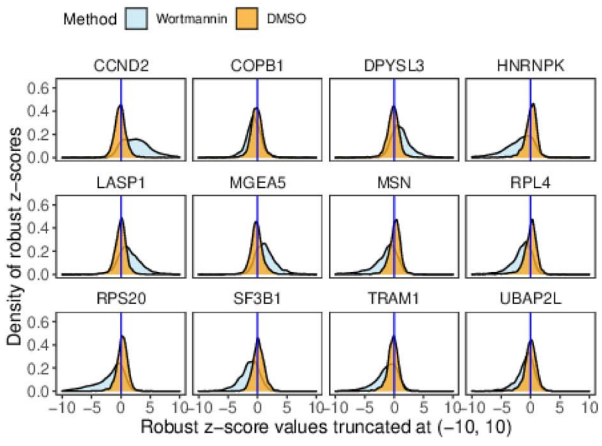
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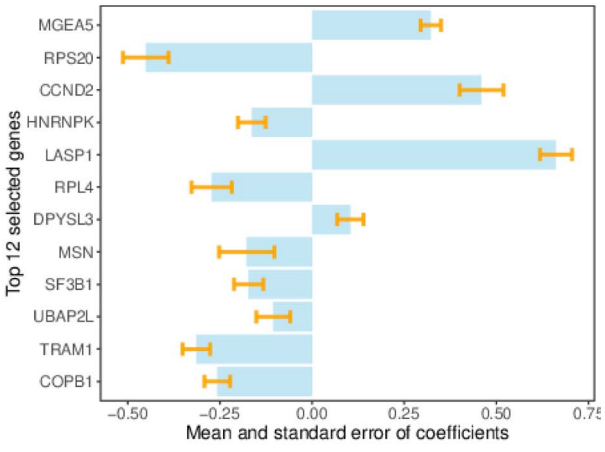
(A) Heatmap of true robust Z-score for selected important genes at FDR=0.1 across 100 replications.



(B) Heatmap of knockoff robust Z-score for selected important genes at FDR=0.1 across 100 replications.



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

Figure 6. Heatmap of true robust Z-scores for selected genes for wortmannin (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

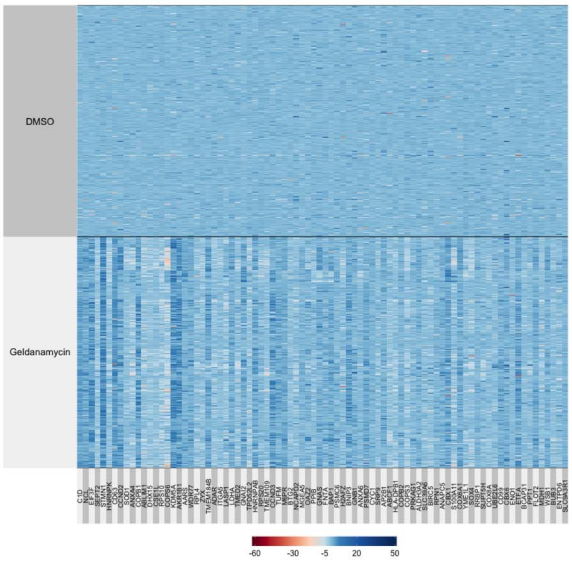
that loss of *UCHL1* expression is a common occurrence with tumor tissues and silencing via methylation of the *UCHL1* promoter has also been observed in biopsies of pancreatic cancer [38] and prostate cancer [39, 40]. The positive coefficient of *UCHL1* (Figure 4) in our analysis is concordant with these previous observations.

Trichostatin-a Our regression analysis produced a positive coefficient (Figure 5) for Cyclin D2 (*CCND2*) which indicates that *CCND2* expression increases with treatment with Trichostatin-a. Takai *et al.* [41] have shown that Histone deacetylase inhibitors (HDACIs) [suberoyl anilide bishydroxamine, valproic acid (VPA), trichostatin A and sodium butyrate] are able to inhibit endometrial cancer cell proliferation and stimulate apoptosis. In the altered gene expressions related to the malignant phenotype, there is a decrease in Cyclin D1 (*CCND1*) and *CCND2*. The positive coefficient of *CCND2* indicates a potential beneficial treatment effect of trichostatin A.

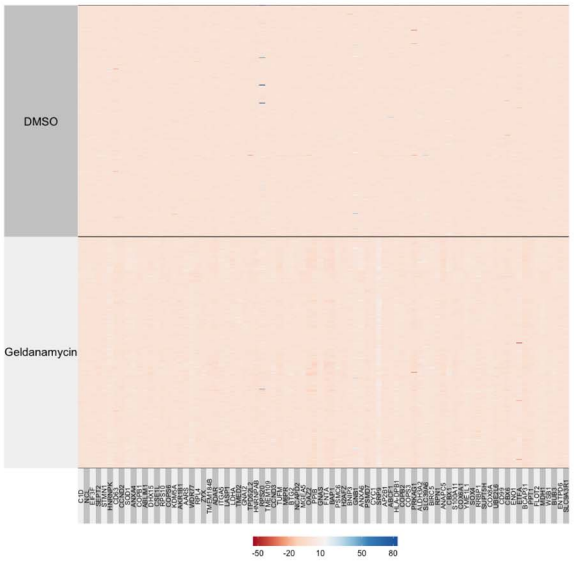
Wortmannin It has been reported that a single nucleotide polymorphism in Meningioma expressed antigen 5 (*MGEA5*), encoding O-Glcnacase (OGA), is associated with type 2 diabetes in a Mexican American population [42]. Wortmannin also has been found to have an effect on reducing insulin signaling and death in seizure-prone mice [43]. In our study, we find that the expression of *MGEA5* exposed to wortmannin perturbations is increased (Figure 6). This leads to the hypothesis that the genetic polymorphism leads to decreased protein activity and treatment by Wortmannin reverses the effects of this decrease by increasing expression levels of the protein.

Geldanamycin Hung *et al.* [44] reported that *CCND2* is a common biomarker in lung and breast cancer. They suggested that increased *CCND2* expression can help inhibit cancer cell growth and migration ability in Taiwanese breast cancer patients. In our study, the positive coefficient (Figure 7) estimate of *CCND2* indicates that the exposure of geldanamycin can help increase

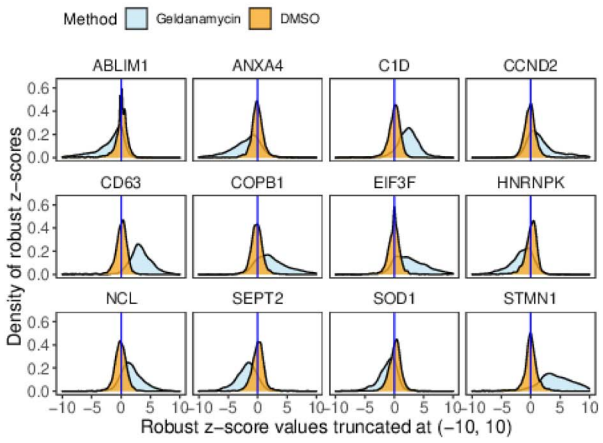
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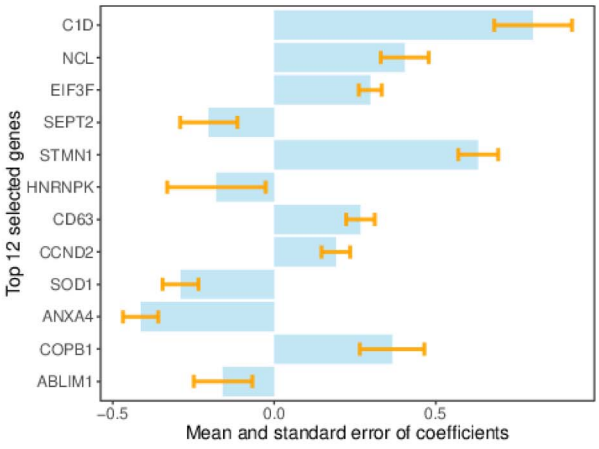
(A)



(B)



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

Figure 7. Heatmap of true robust Z-scores for selected genes for geldanamycin (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

Table 1. Comparison between marginal testing and knockoffs for LINCS data

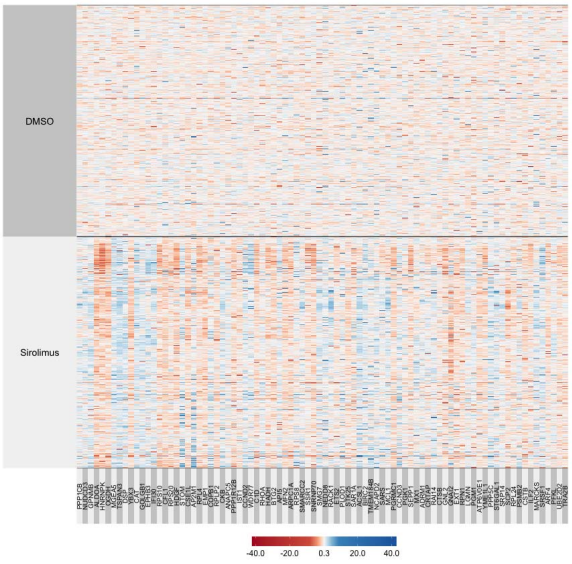
	Marginal only	Knockoffs only	Both
Anthracyclines	0	32	0
Vorinostat	0	54	0
Trichostatin	0	27	0
Wortmanin	0	65	0
Geldanamycin	3	56	12
Sirolimus	2	68	2

expression of *CCND2* and geldanamycin can be a potential drug for breast cancer and lung cancer and is consistent with preliminary reports of its potential [45].

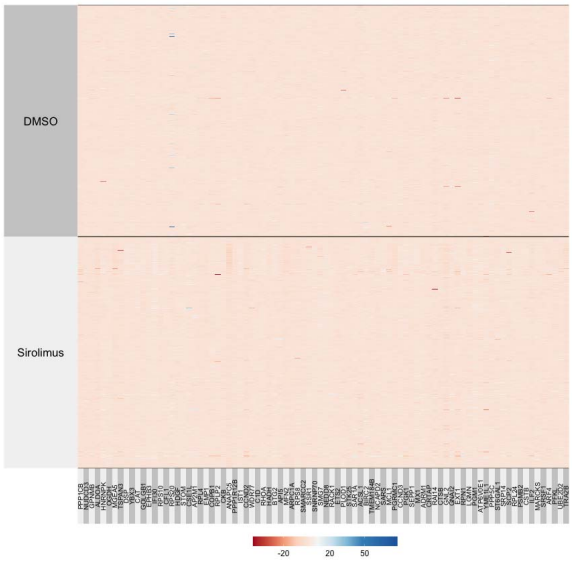
Sirolimus Glycoprotein Nmb (*GPNMB*) is a melanoma-associated glycoprotein that is targeted by the CR011-vcMMAE antibody-drug conjugate and it has been shown that CR011-vcMMAE induces the apoptosis of GPNMB-expressing tumor cells in vitro and tumor regression in xenograft models [46]. That report also suggested the possibility of increasing the anticancer activity of CR011-vcMMAE through pharmacological enhancement of *GPNMB* expression for potential therapeutic benefit. The positive coefficient (Figure 8) of *GPNMB* indicates that sirolimus indeed may have potential therapeutic benefit in combination with other anticancer drugs.

Comparison to marginal testing The standard state-of-the-art methods for identification of important transcriptional responses are based on model-based or model-free marginal testing [47, 48]. Therefore, we compare this approach to our conditional testing approach based on knockoffs and to avoid biases due to model selection we select a model-free permutation testing methodology [47] and use the local FDR method [49] to control the FDR. We use the `coin` R package to test the independence of the perturbation type and the 978 landmark gene expressions [48]. Once we obtain the test statistic value for each gene, we used

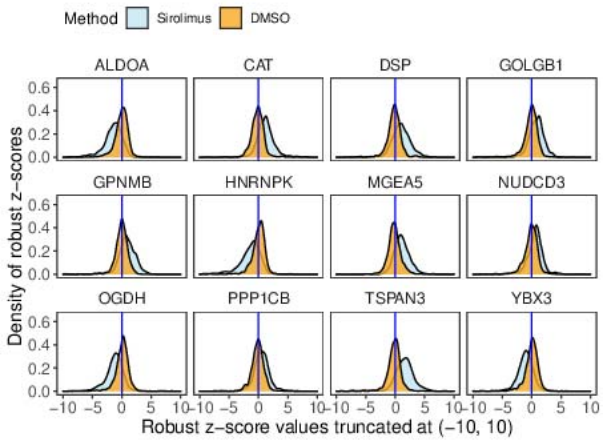
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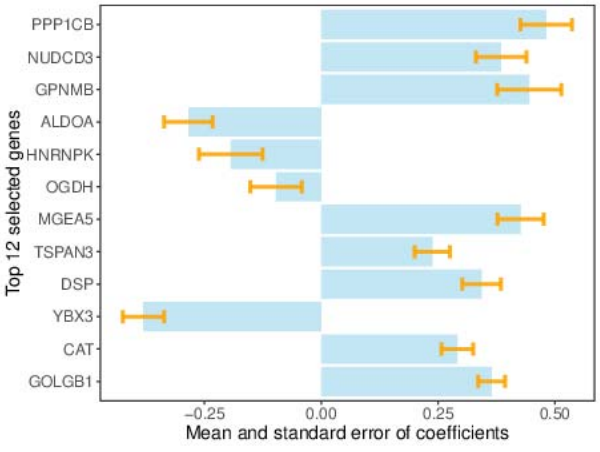
(A) Heatmap of true robust Z-score for selected important genes at FDR=0.1 across 100 replications.



(B) Heatmap of knockoff robust Z-score for selected important genes at FDR=0.1 across 100 replications.



(C) Robust Z-score of top 12 selected genes at FDR=0.1 across 100 replications.



(D) Mean and standard error of coefficients of top 12 selected genes at FDR=0.1 across 100 replications.

Figure 8. Heatmap of true robust Z-scores for selected genes for sirolimus (**A**), knockoff responses (**B**), density (**C**), and bar plots for top selected genes (**D**).

an empirical Bayes technique [49] for large-scale simultaneous hypothesis testing to estimate the local FDR and control the proportion of false positives in the set of genes under the assumption that a majority of the genes modeled are not affected by the perturbations. For each drug, if the estimated FDR for one gene is smaller than the target level of 0.1, the gene will be selected as important. We found that the permutation test method with FDR control at 0.1 is not able to identify any of the genes that were identified by knockoffs for vorinostat, trichostatin-a, wortmannin and anthracycline, respectively. Table 1 shows an overview of the number of genes identified by marginal testing only, knockoffs only and both.

For geldanamycin, permutation test with FDR control selects 15 genes including *SEPT2*, *C1D*, *HNRNPK*, *STMN1*, *RPS10*, *RPS21*, *AKR1B1*, *COPB1*, *PPP2CB*, *TPD52L2*, *DHX15*, *COPS6*, *ADAR*, *KDM5A* and *COPS3*. All the selected genes are also identified by our method except *SEPT2*, *RPS21* and *PPP2CB*. Our method identified 68 out of 978 genes in total under geldanamycin permutations. For sirolimus, permutation test with FDR control only selects *HNRNPK* and *ALDOA*. Both are identified with our method and we identify 70 out of 978 genes in total.

Comparison to sigFeature To compare with another state-of-the-art method, we adapt 'sigFeature' [50] to perform response selection. The method 'sigFeature' aims to find out the significant genes using support vector machine recursive feature elimination method (SVM-RFE) and t-statistics [50] and [51]. 'sigFeature' package is tuned to find differentially significant features along with classification accuracy and works for binary classification. We discuss the results and provide a comparison between our method and 'sigFeature' shown in Figures S11–S16 in the Supplementary Material.

Response Network A comparative analysis of the important genes for each small molecule perturbation (Figure 9) reveals the gene expression changes that are private to each perturbation as well as shared genes. Out of five of all six drug perturbations, *CCND2* and *RPS20* are identified as important gene expressions

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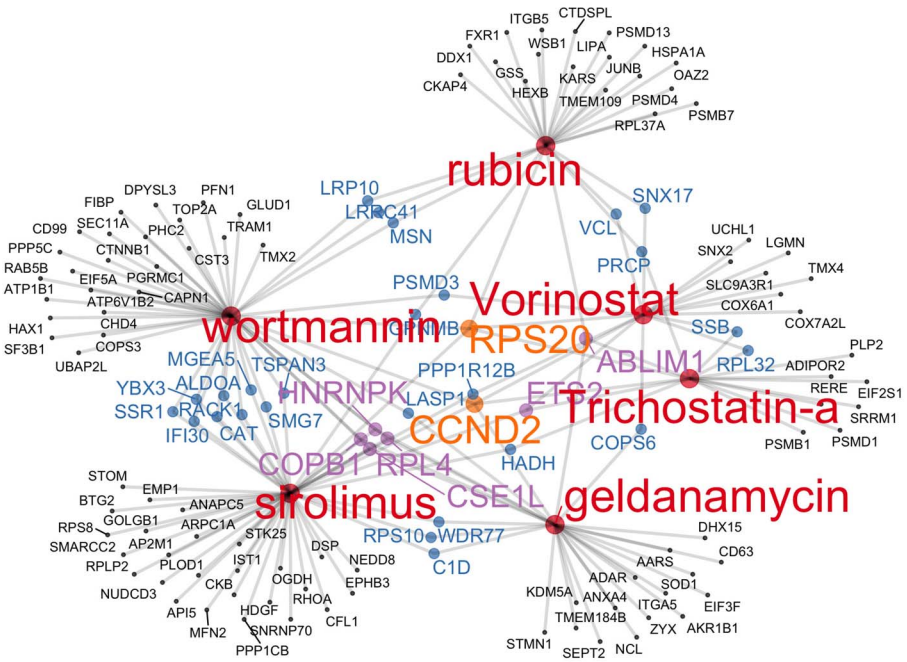


Figure 9. Network plot of identified genes for vorinostat, trichostatin-a, wortmannin, geldanamycin, sirolimus and rubicin. Genes in orange, purple, blue and black represent the common genes identified with exposure to five, three, two and one drug, respectively.

that are affected by the drugs. *CCND2* has been identified as a common therapeutic target in lung and breast cancer [44]. Our identification of the transcriptional response to perturbations by all compounds is interesting in that it may indicate a wider molecular role for this gene. Diseases associated with *RPS20* include familial colorectal cancer type X and diamond-blackfan anemia. *RPS20* shown in Figure 9 has a negative coefficient with all six drugs which indicates a decrease in *RPS20* levels which in turn suggests the potential to contribute to treatment of anti-familial colorectal cancer and diamond-blackfan anemia.

Heterogeneous Nuclear Ribonucleoprotein K (*HNRNPK*) was identified as important and has a negative coefficient under wortmannin, geldanamycin and sirolimus perturbations (Figures 6–8). Poenisch *et al.* [52] reported that an RNA interference of *HNRNPK* results in decreased Hepatitis C virus (HCV) particle production without affecting viral RNA replication. Therefore, our findings suggest that wortmannin, geldanamycin and sirolimus may decreased *HNRNPK* expression and thus decrease HCV particle production. In line with this hypothesis, wortmannin has been suggested as pretreatment drugs for acute liver damage [53].

Human chromosomal segregation 1-like (*CSE1L*) expression is associated with tumor progression in various human cancers. Li *et al.* [54] showed that *CSE1L* was highly expressed in gastric cancer cell lines and *CSE1L* silencing promoted apoptosis and inhibited cell proliferation and invasion. We have found that the the coefficient of *CSE1L* is negative under vorinostat, wortmannin and geldanamycin perturbations (Figures 5, 6, 7), which indicates that vorinostat, wortmannin and geldanamycin are able to decrease expressions of *CSE1L* which may indicate its benefits for gastric cancer treatment.

DISCUSSION

In this paper, we have presented a novel method for controlled response variable selection with theoretical finite sample guarantees on the FDR. We show how the framework can be used in both linear and nonlinear (DNN) models. Synthetic and real data analysis of experimental genomics data sets empirically support the theoretical behavior of the approach. Our analysis of the NIH LINCS data identified important genes whose expression is modulated in response to perturbation with anthracycline, vorinostat, trichostatin-a, geldanamycin and sirolimus, small molecule drugs used in cancer chemotherapy using NIH LINCS data set. Furthermore, we compared the set of important genes that respond to vorinostat, geldanamycin, sirolimus, trichostatin-a, wortmannin and anthracycline to elucidate the response gene network and identify co-responsive pathways. A *potential limitation* is that we only consider a Gaussian distribution of the response variable such that the Y knockoffs can be generated analytically according to our results. Recent work has sought to relax this assumption and in cases where the Gaussian assumption cannot be applied an alternative method for knockoff generation may be more useful [55, 56]. Even so, we have found that the Gaussian assumption practically useful in a real data analysis scenario and the knockoff framework has been shown to be robust to this assumption on F_X [57]. There are several areas that are interesting for future work. In this work, we have assumed the marginal distribution F_Y is known, and this assumption has proven to be reasonable for the data set we analyzed. But in other applications, it may be necessary to learn about the marginal distribution and it would be of interest to offer theoretical bounds on the performance when approximations are employed. In addition, it is also of great interest to design more flexible generation methods to relax the Gaussian assumption for model-Y knockoffs.

Key Points

- Gene expression knockoffs identify important transcriptional changes in response to chemical and genetic perturbations with finite sample control of the false discovery rate.

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- No assumptions on the functional form of the dependence between the gene expression changes and the perturbations.
- Gene expression knockoffs with a deep neural network is used to functionally characterize several uncharacterized small molecule pharmacological agents from the Library of Integrated Network-Based Cellular Signatures (LINCS).

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AUTHORS' CONTRIBUTIONS

T.Z. and P.F. supervised the project and interpreted the results. T.Z. and G.Z. implemented the method and conducted the experiments. G.Z. and T.Z. developed the proofs of the theorems. H.V., G.Z., T.Z. and P.F. wrote and revised the manuscript.

DATA AND CODE AVAILABILITY

The data set GSE92742 is available online (http://lincsportal.ccs.miami.edu/dcic-portal/) and is described in detail by Subramanian *et al.* [1]. This study made use of the level 4 (robust z-score) data. The subset of small molecule perturbations used for this study comprises top six drug classes ranked by number of samples (Vorinostat: 3107; Trichostatin-a: 2586; Wortmannin: 2311; Geldanamycin: 2081; Rubicin: 1937; Sirolimus: 1225) and nine core cell lines. Code to reproduce the experiments is available at https://github.com/flahertylab/deepYknockoff. An R package to apply this protocol is available at https://guang-yu-zhu.github.io/YKnock/.

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